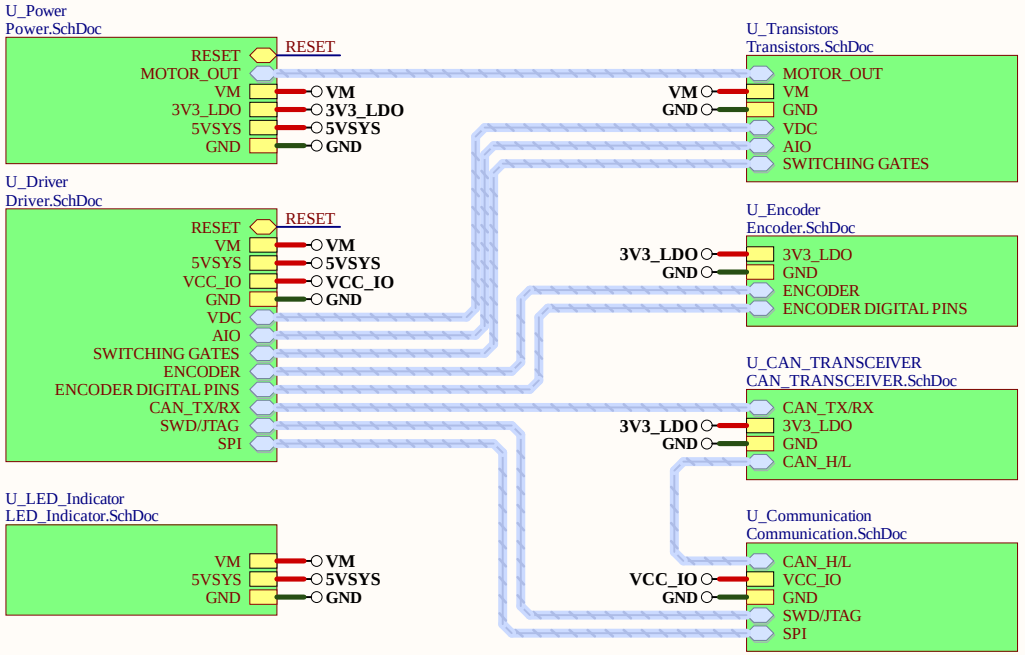


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Title: Top_Sheet.SchDoc	
File: Breakout_Board_PAC5527.PrjPcb	
Date: 2/10/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 8

SUPPORTING CIRCUITRY

VM
C1 220uF
C2 220uF
C3 0.1uF
GND

Connect these capacitors close to the device.

VCC_IO VCC_VCORE VCC_1V8 VP 5VSYS VCC_3V3
C4 2.2uF
C5 4.7uF
C6 22uF
C7 2.2uF
C8 2.2uF
C9 2.2uF
GND

SW1 L1 22uH SW2

CPL C10 0.1uF CPH

VM
VCP C11 1uF

VM
R21 200K RESET

Exposed pad. Must be connected to VSS in a star ground configuration. Connect to a large PCB copper area for power dissipation heat sinking.

GND VCC_VCORE VCC_IO VCC_1V8
PF3 PF4 PF5 PF6 PF7 PE3 PE2 PE1 PE0
49 48 47 46 45 44 43 42 41 40 39 38 37

U1

PF2 1	PF2	PD4	36 MGH
PF1 2	PF1	PD5	35 MGL
PF0 3	PF0	PD6	34 PD6
4	VCC33	PD7	33 PD7
5	VSS	VCP	32 VCP
AIO0 6	AIO0	DRH2	31 WH
AIO1 7	AIO1	DRS2	30 DRSW
AIO2 8	AIO2	DRH1	29 VH
AIO3 9	AIO3	DRS1	28 DRSV
AIO4 10	AIO4	DRH0	27 UH
AIO5 11	AIO5	DRS0	26 DRSU
RESET 12	AIO6	DRL2	25 WL

GND VCC_3V3

GND PF4/ADC4 PF5/ADC5 PF6/ADC6 PF7/ADC7 VCCIO VCORE VCC18 PE3 PE2 PE1 PE0

13 14 15 16 17 18 19 20 21 22 23 24
AIO7 AIO8 AIO9 VSYS VM SW1 SW2 VP CPL CPH DRL0 DRL1
UVDC VVDC WVDC SW1 SW2 CPL CPH UL VL
5VSYS VM VP

PAC5527QMST

DRIVER

REGULATORS

VCC_VCORE = 1.2V
VCC_1V8 = 1.8V
VCC_IO = 3.3V
VCC_3V3 = 3.3V
5VSYS = 5V
VP = 12V

PF6 P7
PF7 P8
PD6 P9
PD7 P10

Communication Testing Points

COMMUNICATION PINS

JTAG: PF0 PF1 PF2 PF3

ENCODER SPI: PE0 PE1 PE2 PE3

ENCODER DIGITAL PINS: PD4 PD5

EXTERNAL SPI: PF4 PF5 PF6 PF7

CAN: PD6 PD7

VCC_VCORE P11
VCC_1V8 P12
VCC_IO P13
VCC_3V3 P14
5VSYS P15
VP P16

Power Testing Points

CAN_TX/RX
PD6 PD6
PD7 PD7
CAN_TX/RX

VM VM
5VSYS 5VSYS
VCC_IO VCC_IO
GND GND

RESET RESET

ENCODER
PE0 PE0
PE1 PE1
PE2 PE2
PE3 PE3
ENCODER

SPI
PF4 PF4
PF5 PF5
PF6 PF6
PF7 PF7
SPI

SWD/JTAG
PF0 PF0
PF1 PF1
PF2 PF2
PF3 PF3
SWD/JTAG

VDC
UVDC UVDC
VVDC VVDC
WVDC WVDC
VDC

AIO
AIO0 AIO0
AIO1 AIO1
AIO2 AIO2
AIO3 AIO3
AIO4 AIO4
AIO5 AIO5
AIO

ENCODER DIGITAL PINS
MGH MGH
MGL MGL
ENCODER DIGITAL PINS

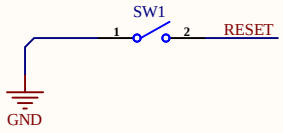
SWITCHING GATES
DRSU DRSU
DRSV DRSV
DRSW DRSW
UH UH
VH VH
WH WH
UL UL
VL VL
WL WL
SWITCHING GATES

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Title: Driver.SchDoc

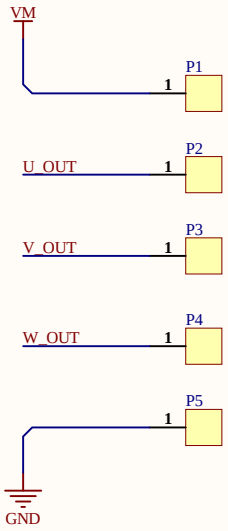
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Date: 2/10/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 8

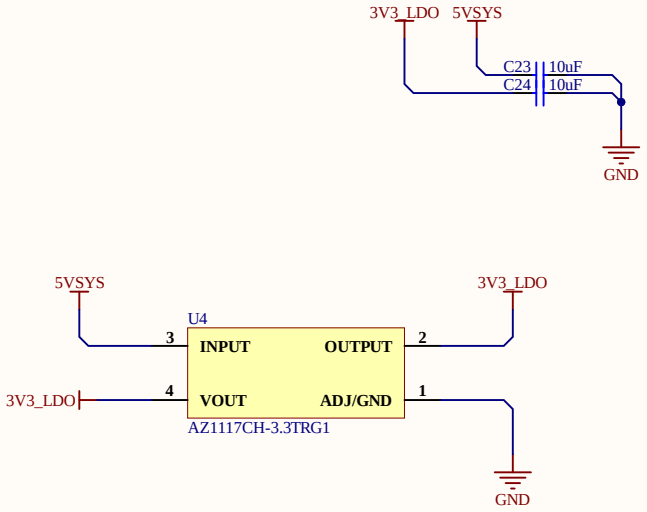


A hard reset of the PAC5527 may be performed by pulling the AIO6 input low for more than 8 seconds when the AIO6 push-button mode is enabled.

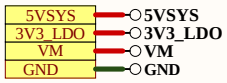
RESET

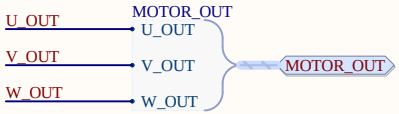


THROUGH HOLE PADS



3V3 REGULATOR

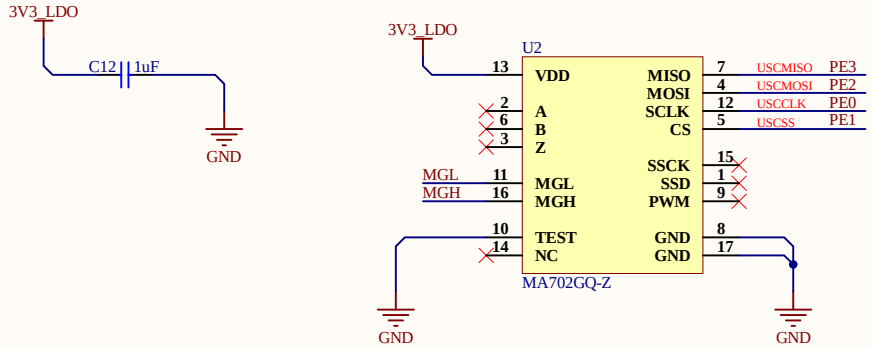


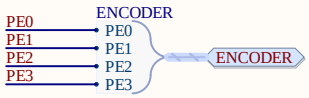


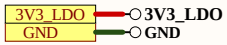


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File: Breakout_Board_PAC5527.PrjPcb	
Date: 2/10/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 8









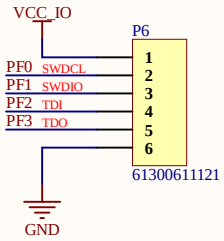
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Title: Encoder.SchDoc

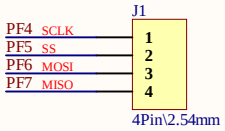
File: Breakout_Board_PAC5527.PrjPcb

Date: 2/10/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 8

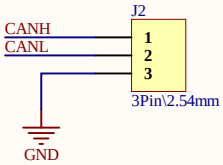
SWD/JTAG

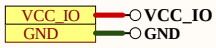


EXTERNAL SPI

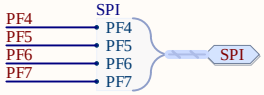


CAN





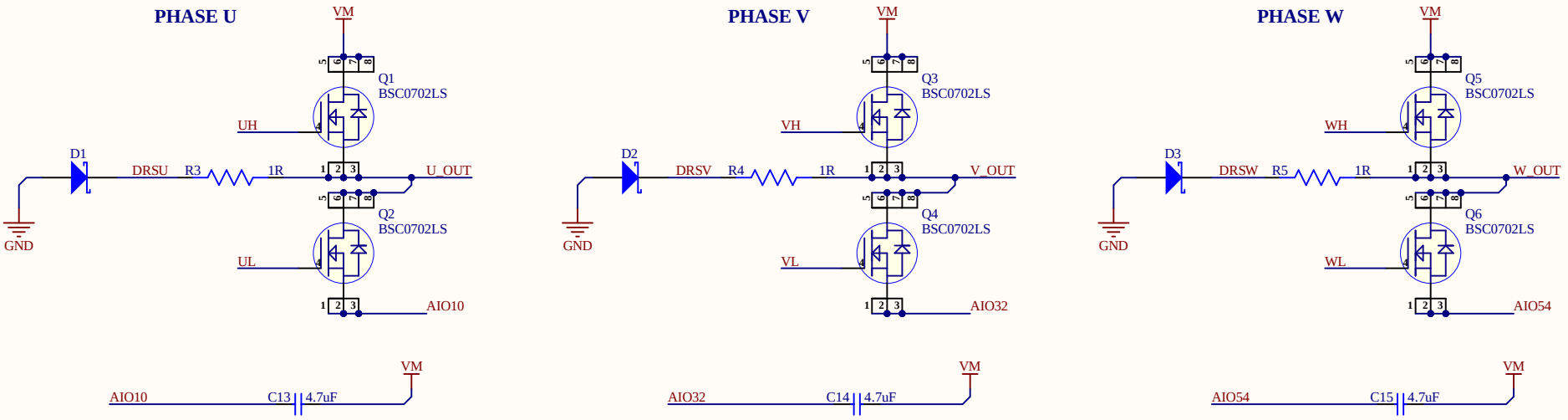


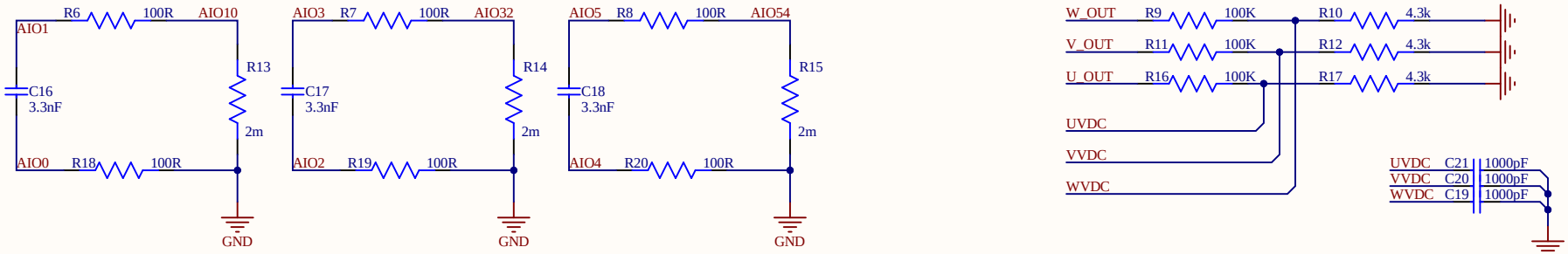




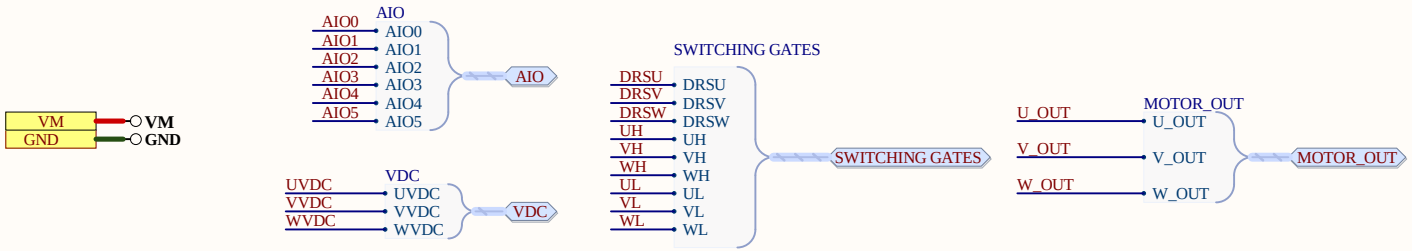
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File does not exist.

Title: Communication.SchDoc	
File: Breakout_Board_PAC5527.PrjPcb	
Date: 2/10/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 8





CONFIGURABLE ANALOG FRONT-END PIN

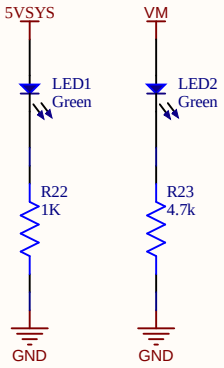


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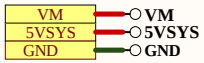
Title: Transistors.SchDoc

File: Breakout_Board_PAC5527.PrjPcb

Date: 2/10/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 8

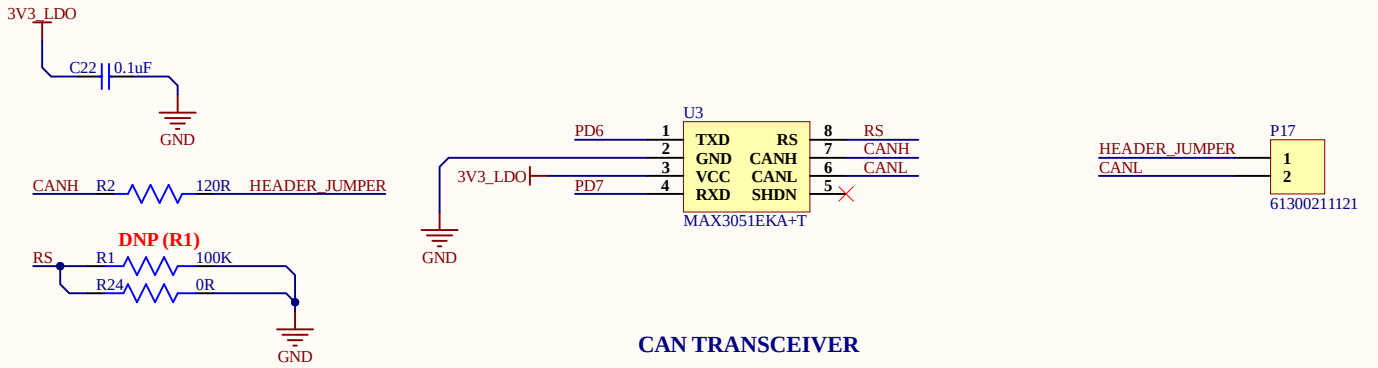


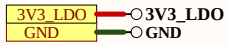
POWER INDICATOR



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Title: LED_Indicator.SchDoc	
File: Breakout_Board_PAC5527.PrjPcb	
Date: 2/10/2023	Page Num: 7
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Title: CAN_TRANSCEIVER.SchDoc	
File: Breakout_Board_PAC5527.PrjPcb	
Date: 2/10/2023	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 8